



ChurchInterpreter

Real-Time Sermon Simultaneous Interpretation Program

User Manual v1.0

Contents

1. Program Overview
2. Installation
3. Issuing API Keys (one-time setup)
4. First Launch — Setup Screen
5. Weekly Usage
6. Screen Layout Guide
7. Frequently Asked Questions

1. Program Overview

ChurchInterpreter translates the preacher's Korean speech into English (or another language) in real time, displaying subtitles simultaneously on the projector and on congregation members' smartphones.

Feature	Description
Real-time interpretation	Translated subtitles appear as soon as the preacher finishes a sentence
Projector display	A dedicated screen with adjustable font size and color
Smartphone access	Congregants view subtitles on their phones via a QR code
Sermon manuscript	Uploading a PDF/Word manuscript improves translation accuracy
Multi-language support	English, Japanese, Chinese, Spanish, and more

2. Installation

1 Run the installer

Double-click `ChurchInterpreter_Setup.exe` to run it.

If a "Windows protected your PC" warning appears, click **More info** → **Run anyway**.

2 Choose the install location

Keep the default path (`C:\Program Files\ChurchInterpreter`) or change it to your preferred location, then click **Install**.

3 Installation complete

A **ChurchInterpreter** icon is created on your desktop.

⚠ The first time you run the program after installation, you must enter your API keys. Read Section 3 below first and have your keys ready.

3. Issuing API Keys (one-time setup)

This program uses two external services. You must obtain a key from each service.

① Anthropic Claude API (for translation)

1 Create an account

Go to <https://console.anthropic.com>

Click **Sign Up** (top right) → enter your email → check the verification email → set a password.

If you already have an account, click **Log In**.

2 Add a payment method

After logging in, click **Billing** in the left menu.

→ **Add payment method** → enter your credit card number, expiration date, and CVC, then save.

→ Click **Buy credits** → enter an amount and pay.

💡 A small amount of \$5–10 is plenty. A once-a-week, one-hour service uses about \$2–5.

3 Create an API key

Click **API Keys** in the left menu.

→ Click **+ Create Key** (top right)

→ Enter a key name (e.g., `church-interpreter`) → click **Create Key**

→ A key starting with `sk-ant-api03-...` appears — **be sure to copy it** to a safe note (you cannot view it again after closing the window).

⚠ The API key is like a password. Do not share it or send it by email. If it is lost, delete the old key and issue a new one.

② Google Cloud Speech-to-Text key

1 Create a Google Cloud project

Go to <https://console.cloud.google.com> (log in with your Google account).

→ Click the project dropdown in the top bar (shows "Select a project" or an existing project name).

→ Click **New Project** at the top right.

→ Enter a project name (e.g., `church-interpreter`) → click **Create**.

→ Once created, confirm the project is selected.

2 Link a billing account

Top-left menu (☰) → click **Billing**.

→ **Link a billing account** → register a credit card.

(Free if monthly usage stays within the limit — up to 60 minutes of speech recognition per month can fall within the free tier.)

3 Enable the Speech-to-Text API

In the top search bar, type **Cloud Speech-to-Text API** and click the result.

→ Click the blue **Enable** button (if already enabled, it shows "API enabled").

4 Create a service account

Top-left menu (☰) → **IAM & Admin** → click **Service Accounts**.

→ Click **+ Create Service Account** at the top.

→ Enter a service account name (e.g., `church-stt`) → click **Create and Continue**.

→ Select role: search **Cloud Speech Client** and select it → **Continue** → **Done**.

5 Download the JSON key file

In the service account list, click the account you just created (`church-stt@...`).

→ Go to the **Keys** tab.

→ **Add Key** → **Create new key**.

→ Choose key type **JSON** → click **Create**.

→ A `.json` file downloads automatically — keep this file safe.

⚠ The JSON key file is like a password. Do not email it or share it externally. If it is lost, delete the old key and issue a new one.

③ ngrok account (for smartphone access)

1 Sign up for ngrok

<https://dashboard.ngrok.com> → Sign Up (free)

2 Copy your authtoken

After logging in, copy the token from **Getting Started → Your Authtoken**.

3 (Optional) Get a fixed domain

Domains menu → **New Domain**

Issue one free fixed domain (lets you reuse the same QR code every week).

4. First Launch — Setup Screen

1 Run the program

Double-click **ChurchInterpreter** on the desktop.

A black console window opens and the server starts. (Do not close it.)

2 Open the setup screen

A browser opens automatically, or type this in the address bar:

```
http://localhost:5000/setup
```

3 Enter your Claude API key

Enter the `sk-ant-...` key you copied in Section 3, then click **Test Connection**.

4 Upload the Google key file

Use the **Choose File** button to select the downloaded `.json` file, then click **Test Connection**.

5 Configure ngrok

Enter your authtoken, and your fixed domain if you have one.

6 Save settings

Click **Save Settings** → click **Open Operator Dashboard**.

✓ Settings are saved to the `.env` file in the program folder and apply automatically on subsequent runs.

5. Weekly Usage

When	Task	How
Before service	Run the program	Double-click ChurchInterpreter on the desktop
Before service	Open the operator dashboard	<code>http://localhost:5000/operator</code>
Before service	Upload the sermon manuscript (optional)	"Manuscript Upload" on the left of the dashboard → select PDF/Word
Before service	Open the projector display	<code>http://localhost:5000</code> → click Fullscreen
Before service	Guide congregants with the QR code	Print the dashboard QR code or show it on screen
Service start	Turn the mic on	Click the "🎤 Mic On" button on the dashboard
During service	Check subtitles	Watch real-time subtitles on the projector and on phones
After service	Turn the mic off	Click the "🔇 Mic Off" button on the dashboard
After service	Close the program	Close the console window or press Ctrl+C

6. Screen Layout Guide

Operator Dashboard (/operator)

- **Mic On/Off** — start / stop interpretation
- **Language selection** — change the source / target language
- **Manuscript upload** — improves translation accuracy with a PDF or Word file
- **Display settings** — adjust font size, color, and background (applied to the projector in real time)
- **QR code** — the connection address for congregants' phones

Projector Screen (/)

- Double-click or F11 — toggle fullscreen

- Subtitles stack from the bottom upward; when the screen is full, the oldest lines disappear from the top

Congregant Phone Screen (/mobile)

- Connect by scanning the QR code
- Includes a keep-awake feature so the screen does not turn off

7. Frequently Asked Questions

Q. Translation is too slow.

Depending on your internet speed and the API server status, a delay of 0.5–3 seconds can occur. A wired internet connection is recommended.

Q. A proper noun (a person's name or a Bible place name) is recognized incorrectly.

On the dashboard, go to **Glossary Upload** and add church-specific proper nouns to improve recognition accuracy.

Q. I scanned the QR code on my phone but can't connect.

Check that ngrok setup is complete. The address should be shown in the QR code area of the dashboard.

Q. I accidentally closed the black console window.

Double-click the desktop icon again to restart the program.

Q. I want to change my API key.

Go to `http://localhost:5000/setup` in your browser, enter the new key, and save.

Q. How much does translation cost?

About \$2–5 for a once-a-week, one-hour service. You can check real-time usage in the Anthropic console.